Cover letter

Linear Algebra and its Applications 25 October 2026

Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript “Nilpotent-free spaces of 3 by 3 matrices over the field with three elements” by Deep Bhattacharjee.

The paper proves that the largest nilpotent-free subspace dimension in 3 by 3 matrices over the field with three elements is five. A direct algebraic construction gives the lower bound, while a finite reduction and independently checked search establish the upper bound and refute the proposed six-dimensional construction in this case.

The exact extremal dimension of a finite-field matrix space, its explicit construction and structural analysis fit finite-dimensional matrix theory and its arithmetic and combinatorial aspects.

The prior public preprint is https://hexagonmath.org/2610.00480v1. The research manuscript/source archive is https://doi.org/10.5281/zenodo.23607867. Editable sources and verification records are supplied with the manuscript.

The Lean development formalises the reduction and search soundness. Two compiled evaluations remain explicit parts of its trust base. This must not be described as an audit using only the three standard axioms.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The submitting author has declared that all parts have been checked and understood by a human. The named human authors retain responsibility for the content.

Sincerely,

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